

The Impact of an Integrated Outpatient Specialty Neurology Pharmacist in a Non-MS Clinical Setting

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Code

VANDERBILT UNIVERSITY  
MEDICAL CENTER

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CONCLUSION

Over 3 months, outpatient specialty neurology pharmacists performed **2,244 interventions in 741 patients** that often resulted in a recommendation for therapy change or additional monitoring

Pharmacists avoided **direct costs of up to almost \$200,000**

OBJECTIVE

To examine the role of an integrated outpatient specialty neurology pharmacist in the non-MS setting by evaluating interventions performed

METHODS

- Setting
  - Outpatient non-MS neurology clinics (ATTR amyloidosis, Autonomics, Epilepsy, Movement disorders, Neuromuscular, Others) at an academic medical center with an integrated specialty pharmacy
- Design
  - Single-center, retrospective cohort study
    - January 2023 – March 2023
- Sample
  - All patients prescribed a specialty medication from the non-MS Neurology clinics at Vanderbilt University Medical Center with at least one pharmacist intervention performed during the study period
- Data Procedures
  - Data was collected from the electronic health record and specialty pharmacy management system
  - Interventions documented by the pharmacist during normal clinical practice were extracted and reviewed
- Analysis
  - Cost avoidance: estimated by calculating the total costs of medications associated with a follow-up outcome of discontinuation, medication change, or dose change
  - Ordinal mixed effects regression model for the factors associated with greater intervention time
    - Covariates: filling pharmacy, clinic, age, and prescription insurance

Primary Outcome

- Number of interventions resulting in a recommendation made to a provider or patient
- Time pharmacist spent on interventions

Secondary Outcomes

- Outcome of recommendations
- Intervention impact score\*
- Cost avoidance
- Factors associated with greater intervention times
- Number of recommendations accepted

\*Impact score definition

- review only, no intervention needed
- quality of life impact, intervention needed
- negative impact on health, intervention needed
- ED, hospitalization, or death likely, intervention needed

RESULTS

Table 1: Baseline Demographics (N=741)

|                                        |            |
|----------------------------------------|------------|
| Age, years, median (IQR)               | 60 (22-72) |
| Sex, male, n (%)                       | 406 (55)   |
| Race, White, n (%)                     | 599 (83)   |
| Primary Pharmacy Insurance, n (%)      |            |
| Medicare                               | 340 (46)   |
| Commercial                             | 221 (30)   |
| Medicaid                               | 132 (18)   |
| Tricare/Other                          | 30 (4)     |
| None                                   | 18 (2)     |
| Integrated Pharmacy Status, yes, n (%) | 511 (69)   |

IQR: interquartile range

Figure 1: Pharmacist Interventions Resulting in Recommendations

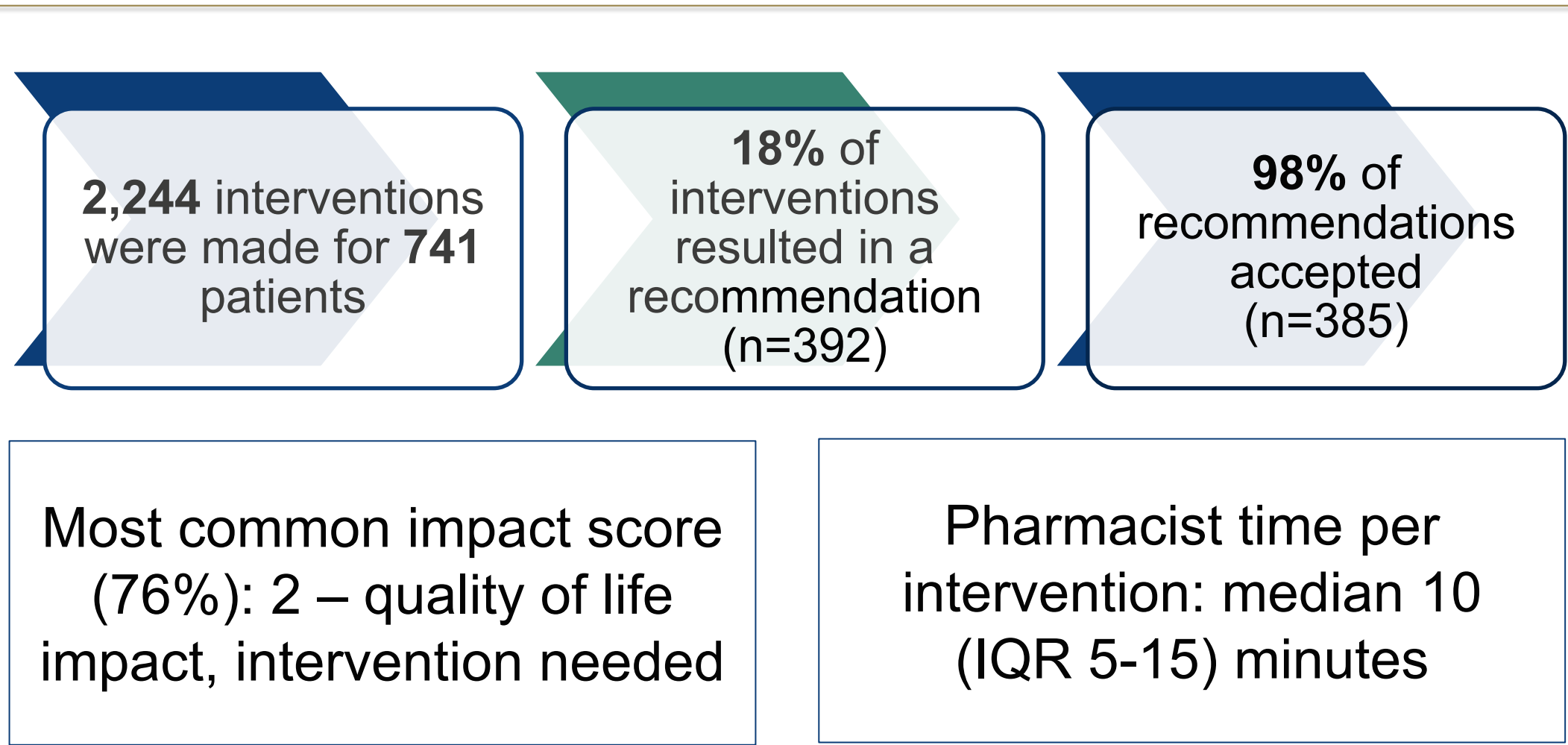
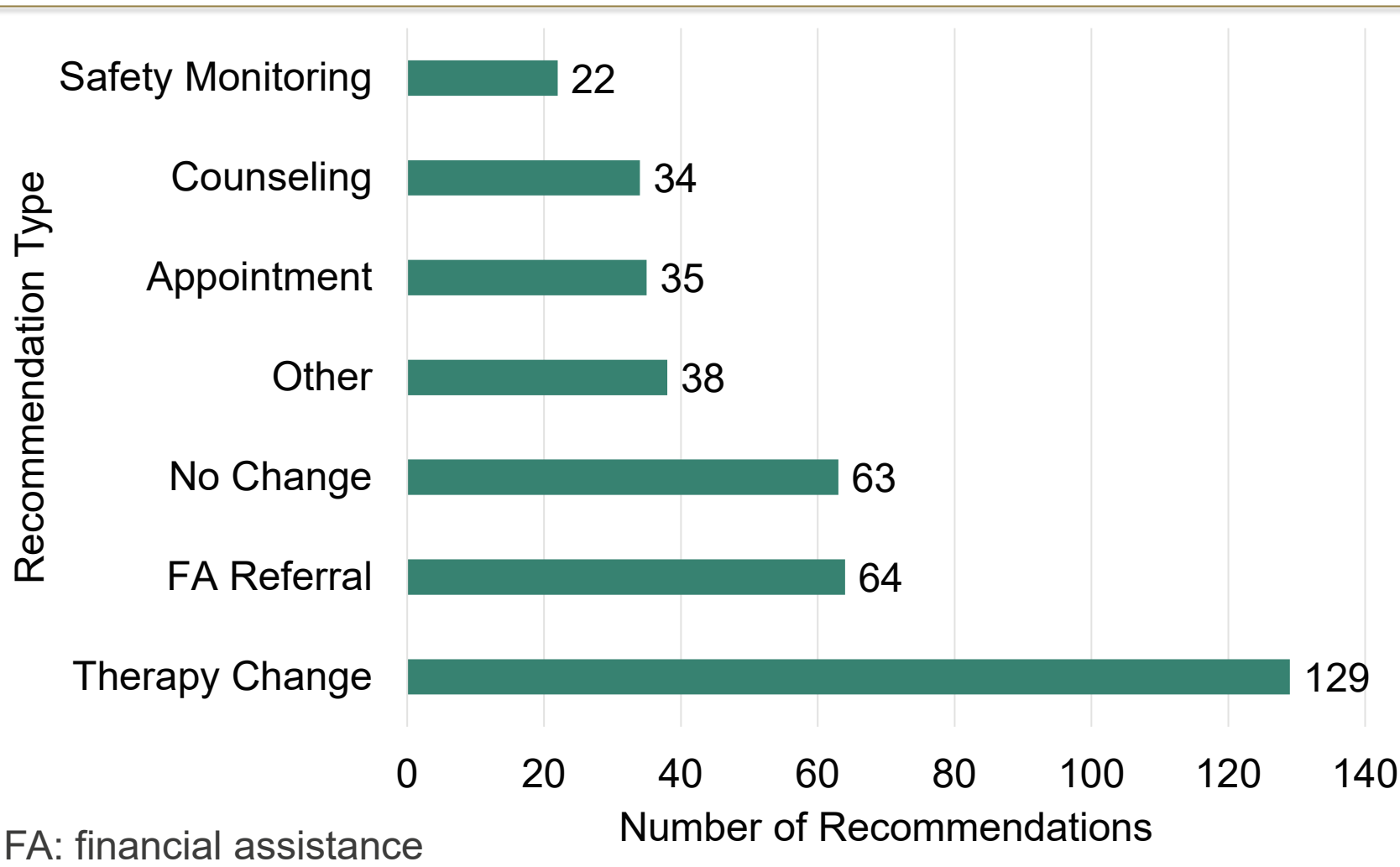
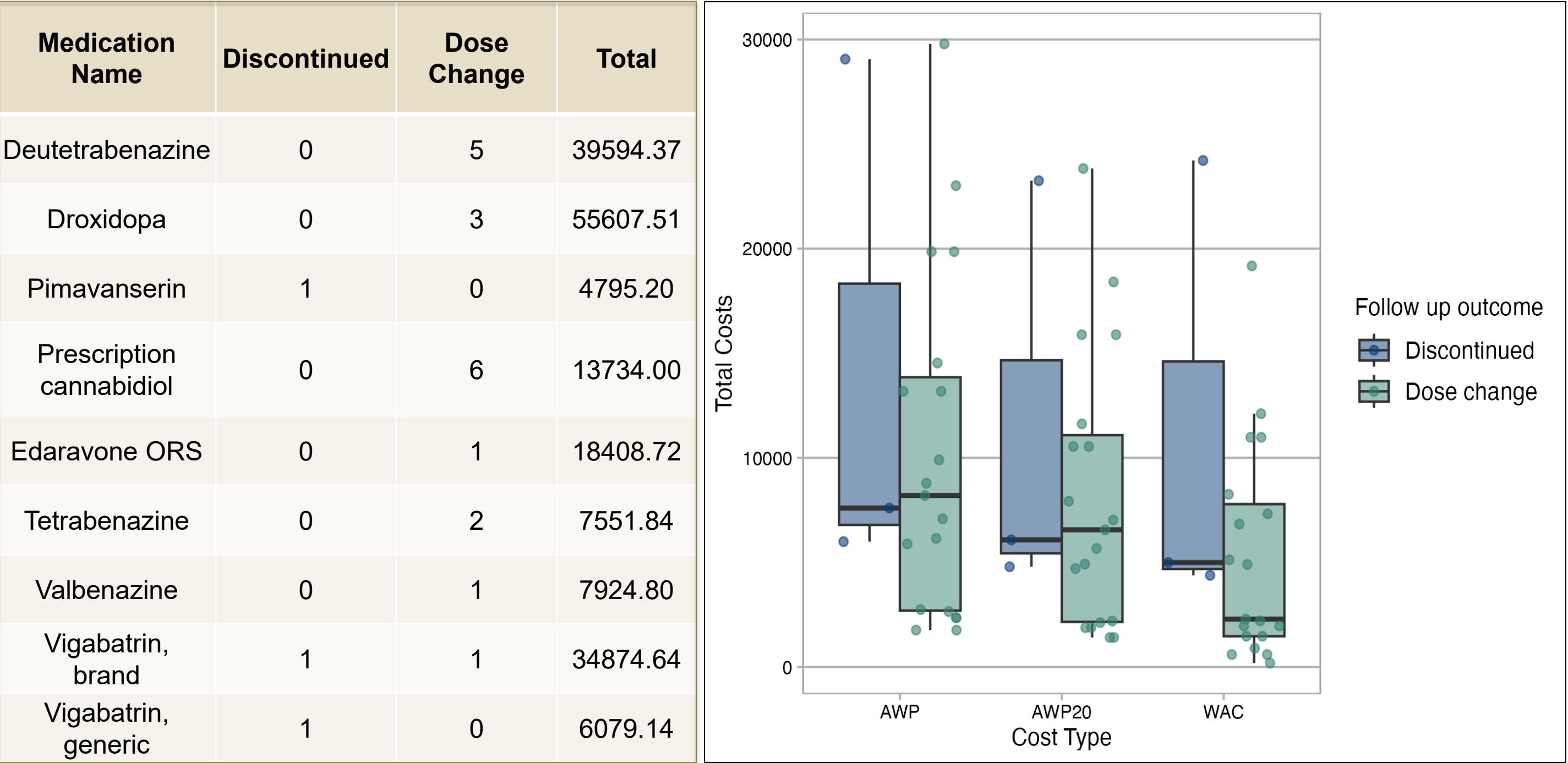


Figure 2: Pharmacist Recommendation Outcome



FA: financial assistance

Figure 3 and Table 2: Cost Avoidance Due to Recommendations and by Medication



Cost avoidance totaled **\$188,570** using the average wholesale price minus 20%  
Dose changes resulted in **\$154,446** in cost savings while discontinuations totaled **\$34,124**

AWP: average wholesale price; AWP20: AWP – 20%; WAC: wholesale acquisition cost

Figure 4: Factors Associated with Greater Intervention Time

| Ordinal mixed effects regression model - Odds Ratio |            |         |         |        |               |
|-----------------------------------------------------|------------|---------|---------|--------|---------------|
| Confidence Interval                                 |            |         |         |        |               |
| Variable                                            | Odds Ratio | LowerCI | UpperCI | PValue | Visualization |
| VSP Patient (Ref = No)                              |            |         |         |        |               |
| Yes                                                 | 0.856      | 0.718   | 1.018   | 0.079  | ↓             |
| Clinic (Ref = Movement Disorders)                   |            |         |         |        |               |
| Amyloidosis                                         | 0.696      | 0.487   | 0.995   | 0.047  | ↓             |
| Autonomics                                          | 1.259      | 0.998   | 1.587   | 0.052  | ↑             |
| Epilepsy                                            | 0.734      | 0.544   | 0.990   | 0.043  | ↓             |
| Inpatient                                           | 0.672      | 0.486   | 0.927   | 0.015  | ↓             |
| Neuromuscular                                       | 1.267      | 0.972   | 1.652   | 0.079  | ↑             |
| Other                                               | 3.959      | 0.899   | 17.427  | 0.069  | ↑             |
| Age                                                 |            |         |         |        |               |
| Years                                               | 1.001      | 0.995   | 1.006   | 0.833  | ↔             |
| Rx Insurance (Ref = Commercial)                     |            |         |         |        |               |
| Medicaid                                            | 0.829      | 0.636   | 1.081   | 0.166  | ↓             |
| Medicare                                            | 0.820      | 0.656   | 1.024   | 0.08   | ↓             |
| None                                                | 1.385      | 0.807   | 2.382   | 0.238  | ↑             |
| Tricare or Other                                    | 0.848      | 0.569   | 1.262   | 0.417  | ↓             |



**Clinic specialty** was associated with longer intervention time (p < 0.001)

Table 3: Medications with an Intervention

| Medication Name                                 | N=1034 % (n) |
|-------------------------------------------------|--------------|
| Amantadine extended-release capsule             | 3.2 (33)     |
| Apomorphine sublingual                          | 0.7 (7)      |
| Deutetrabenazine                                | 16.4 (170)   |
| Droxidopa                                       | 21.6 (223)   |
| Edaravone ORS                                   | 2.9 (30)     |
| Everolimus                                      | 0.7 (7)      |
| Immune globulin subcutaneous [human] 20% liquid | 1.8 (19)     |
| Inotersen                                       | 0.2 (2)      |
| Istradefylline                                  | 1.1 (11)     |
| Pimavanserin                                    | 4.8 (50)     |
| Prescription cannabidiol                        | 21.5 (222)   |
| Riluzole                                        | 2.6 (27)     |
| Sodium Phenylbutyrate/ursodoxicoltaurine        | 0.7 (7)      |
| Tafamidis                                       | 4.4 (46)     |
| Tetrabenazine                                   | 4.3 (44)     |
| Valbenazine                                     | 6.3 (65)     |
| Vigabatrin                                      | 1.9 (20)     |
| Other (non-specialty medication)                | 4.9 (51)     |

Total number of prescriptions with at least one intervention  
Patients who have more than one dose are counted multiple times

**Disclosures:** authors have nothing to disclose